

Dimitry Grebenyuk

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Shenzhen, China | dimitrygrebenyuk@icloud.com | +86-13530042782 | github.com/grebenyyk | ORCID

Research Interests

Lanthanide coordination chemist and materials scientist (PhD, Lomonosov Moscow State University), currently Senior Lecturer at Shenzhen MSU-BIT University. My research focuses on developing rare-earth metal–organic frameworks and polynuclear clusters, combining conventional crystallography with total-scattering analysis and machine learning–driven structural modeling (DFT, molecular simulations) to resolve their structural complexity. I lead inorganic chemistry teaching and supervise research on functional materials.

Education

Lomonosov Moscow State University , PhD, Inorganic Chemistry Thesis: «Coordination polymers and polynuclear clusters based on aliphatic rare earth carboxylates»	2018 – 2022
Lomonosov Moscow State University , MSc <i>cum laude</i> , Chemistry	2012 – 2018

Academic Appointments

Senior Lecturer , Faculty of Materials Science, Shenzhen MSU-BIT University Instructor and research supervisor in chemistry and materials science (previously Teaching Assistant, 2019–2023).	2023 – Present
Doctoral Researcher & Research Engineer , Department of Chemistry, Lomonosov Moscow State University Doctoral research on lanthanide coordination polymers; concurrent diffraction-support engineering role.	2018 – 2023

Selected Publications

- **Grebenyuk, D.**, Sabitova, I., Long, H., Gavrilkin, S., Tsvetkov, A., Lyssenko, K., Tsymbarenko, D. Solid-state transformation and structural diversity in lanthanide 1D coordination polymers with bulky cyclohexanecarboxylate ligands. *J. Solid State Chem.* **2026**, 360, 126085.
- **Grebenyuk, D.**, Shaulskaya, M., Shevchenko, A., Zobel, M., Tedeeva, M., Kustov, A., Sadykov, I., Tsymbarenko, D. Tuning the Cerium-Based Metal–Organic Framework Formation by Template Effect and Precursor Selection. *ACS Omega* **2023**, 8 (50), 48394–48404.
- **Grebenyuk, D.**, Zobel, M., Tsymbarenko, D. Partially Ordered Lanthanide Carboxylates with a Highly Adaptable 1D Polymeric Structure. *Polymers* **2022**, 14 (16), 3328.
- Tsymbarenko, D., **Grebenyuk, D.**, Burlakova, M., Zobel, M. Quick and robust PDF data acquisition using a laboratory single-crystal X-ray diffractometer for study of polynuclear lanthanide complexes in solid form and in solution. *J. Appl. Cryst.* **2022**, 55, 890–900.
- **Grebenyuk, D.**, Zobel, M., Polentarutti, M., Ungur, L., Kendin, M., Zakharov, K., Degtyarenko, P., Vasiliev, A., Tsymbarenko, D. A Family of Lanthanide Hydroxo Carboxylates with 1D Polymeric Topology and Ln₄ Butterfly Core Exhibits Switchable Supramolecular Arrangement. *Inorg. Chem.* **2021**, 60 (11), 8049–8061.

Full publication list (h-index 7, ~200 citations): scholar.google.com

Teaching & Supervision

Courses developed and taught

- Designed and delivered the undergraduate course *Introduction to Chemistry*; authored the accompanying textbook, published by MSU-BIT.
- Teach full undergraduate courses: *General Chemistry and Chemistry of Elements* (seminars and laboratory sessions) and *Modern Inorganic Chemistry* (seminars).
- Deliver an annual lecture on the *History of Materials Science* for master's students.

Student supervision

- Supervised 23 bachelor's and master's thesis projects on metal–organic frameworks and luminescent materials; 16 students have continued to graduate programs.
- Alina Zhumabekova's thesis «Sorption of molecular iodine in lanthanide terephthalate metal–organic frameworks» was commended by the State Degree Committee overseeing the defense.

Recognition: Second Prize, MSU-BIT University Competition for Pedagogical Innovations (2025), as a team member.

Invited Talks & Conferences

- **Beijing Conference on Molecular Sciences**, 2025, Beijing — “*Molecular clusters of f-elements: from synthesis to deep learning insights*” (invited)
- **Peking University International Forum on “AI + New Materials”**, 2024, Shenzhen — “*Deep learning insight into molecular clusters nuclearity via pair distribution function*” (invited)

Professional Service

Peer reviewer for *Frontiers in Nuclear Engineering*.

Software & Data

PDF-NN — neural network for predicting the nuclearity of lanthanide clusters from pair distribution function (PDF) data. Source code, trained models, and accompanying dataset publicly released under the MIT license.

Code & models: github.com/grebenyyk/PDF-NN Data: Zenodo, DOI 10.5281/zenodo.19173873

OEIS A000602 — contributed the higher-order correction terms in the asymptotic formula for the number of alkane structural isomers, accepted into the OEIS entry.

Entry: oeis.org/A000602 Code & write-up: github.com/grebenyyk/alkane-isomers

OEIS A397444 — derived the closed-form expression for the anion excess of saturated rock-salt (NaCl) nanocubes, $a(n) = 3n^2 + 1 - (-1)^n$, as an exact surface-stoichiometry descriptor for ionic nanocrystals.

Entry: oeis.org/A397444 Code & write-up: github.com/grebenyyk/np-stoichiometry

Skills & Expertise

Research: Lanthanide coordination chemistry, Metal–organic frameworks, Crystallography (SC-XRD, PXRD), Total scattering and PDF analysis, Synchrotron beamtime (Diamond, Elettra, Kurchatov)

Computing: Python, Machine learning for structure prediction from PDF data, DFT & molecular simulations (VASP, Quantum Espresso, RASPA), High-performance computing, Crystallographic software (SHELX, Olex2)

Languages: Russian (native), English (fluent), Mandarin Chinese (HSK4)